MITOCHONDRIAL & CELLULAR ENERGY

NAD+ — the full review

CELLULAR ENERGY COENZYME

This is the deeper look behind the short version — what the human trials of injected NAD+ actually show, why infusion speed matters, and what your consultation covers.

What it is

NAD+ (nicotinamide adenine dinucleotide) is not a peptide or a novel compound — it is a coenzyme every cell already makes and uses constantly, central to how a cell converts food into usable energy and required by enzymes that repair damaged DNA and regulate cell-aging processes. The body builds it from dietary vitamin B3 and related precursor molecules, as part of that vitamin's core biological job.

What is offered clinically is the injected or IV form — a compounded preparation held to a different regulatory standard than an over-the-counter oral supplement such as nicotinamide riboside or nicotinamide mononucleotide. NAD+ levels measured in some tissues do appear to fall with age and with poor sleep, alcohol use, and chronic illness, which is the basis of a large, genuinely active field of aging biology — but a large 2026 analysis of blood NAD+ across seven cohorts found the molecule "surprisingly stable" and did not consistently fall with age the way the popular narrative assumes, a newer, unsettled finding that complicates the simple "NAD+ declines, so replace it" premise.

Human trials of the injected form are limited, and infusion speed matters — going too fast is the most common reason people feel uncomfortable during a session, which is why infusions are typically given slowly, over several hours.

What people use it for

USED FOR	WHAT THAT MEANS
Energy and general vitality	Solid underlying cell biology, but few controlled human trials exist of the injected form specifically.
Cognitive clarity	One small peer-reviewed trial reported a benefit — a real but genuinely limited, unreplicated finding.
Anti-aging and longevity	No controlled human trial of the injected form has tested this outcome, and the underlying premise that NAD+ reliably declines with age has itself been challenged by newer research.
Addiction and withdrawal support	The most substantial human dataset for injected NAD+ of any use, but built on an uncontrolled, multi-ingredient chart review with low follow-up response.
Exercise performance or recovery	No controlled human trial of IV or injected NAD+ for this use has been identified; promotional material traces to clinic marketing, not published research.

How it works

NAD+ and its reduced partner, NADH, form the cell's central electron shuttle: in every mitochondrion, this pair cycles constantly to move electrons through the chain that generates

ATP, the cell's energy currency. NAD+ is also consumed as a substrate by two enzyme families tied to DNA repair and cell regulation — sirtuins, studied heavily in longevity research, and PARPs, which detect and help repair damaged DNA, using up NAD+ each time they act.

Chronic DNA damage or metabolic stress can drive PARP activity up, one proposed mechanism for age-or stress-related decline in some tissues.

The gap between this real biology and the infusion claim is where the marketing gets ahead of the evidence. NAD+ is a large, charged molecule not expected to cross an intact cell membrane on its own, and no clear, established transporter moves whole extracellular NAD+ into human cells at meaningful scale; a small pilot pharmacokinetic study found plasma clearance patterns consistent with NAD+ being broken down outside cells before it would ever reach the inside of one. A 2026 randomized pilot trial found that IV nicotinamide riboside, a precursor, reliably raised measured blood NAD+, while IV NAD+ itself, unexpectedly, did not show the same clear elevation — a finding that directly cuts against the logic used to justify infusing the finished molecule rather than a precursor.

What the research actually looked at

FDA's own review of nominated indications located only five qualifying human studies of injected or oral NAD across nearly six decades, from 1961 to 2019: 124 patients given IV NAD for addiction, 12 given oral NADH for chronic fatigue, and a handful more for Parkinson's disease — a genuinely small evidence base for a widely marketed treatment.

The most-cited human data, for addiction and withdrawal support, comes from a retrospective chart review of about 60 patients given IV NAD+ as one part of a multi-ingredient protocol — not randomized, not blinded, with no control group, and only 45% survey follow-up at 12 to 20 months. A statistically significant drop in withdrawal scores by day 5 was reported, but because NAD+ was one part of a larger protocol, its individual contribution was never isolated.

A 2026 systematic review of 113 NAD+ studies found zero controlled human trials of IV NAD+ for anti-aging purposes. One small, peer-reviewed study reported IV NAD+ improved measures of cognitive performance in human subjects — a genuine but single, unreplicated finding that should be read as preliminary rather than established.

Common misunderstandings

> “IV NAD+ is proven to slow or reverse aging.” No controlled human trial of IV NAD+ has ever tested an anti-aging outcome — a 2026 systematic review of 113 NAD+ studies found zero. The premise that NAD+ reliably declines with age was itself challenged by a large 2026 cross-cohort blood analysis finding levels "surprisingly stable."
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> “It's basically a vitamin-related molecule, so it's safe at any dose or infusion speed.” FDA's own review specifically flags that IV NAD+ must be given slowly, over 6 to 8 hours, because a fast infusion produces chest tightness, stomach cramping, headache, and shortness of breath; separately, FDA has received adverse event reports of severe chills, shaking, vomiting, and fatigue tied to contaminated product.
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> “The oral NAD+ research proves the IV version works, since it's the same molecule.” Oral precursor research and IV NAD+ research are different literatures studying different products, bioavailability, and endpoints. The mechanism used to justify IV NAD+ — direct cellular restocking — is itself in question: NAD+ is a large polar molecule not expected to cross cell membranes intact, and a 2026 pilot trial found IV NR raised measured blood NAD+ more reliably than IV NAD+ itself did.
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> “It's a prescription-grade, quality-controlled treatment because a clinic administers it.” FDA has directly warned that some compounders have used food-grade NAD+, material intended for supplements rather than sterile injection, to make IV products

— and has documented real consequences, including a warning letter after patients developed adverse reactions to product later found to contain high bacterial endotoxin levels.

Who it is not for

- Anyone unable to tolerate a slow, multi-hour infusion — going too fast is the most common reason people experience chest tightness, cramping, or headache during a session.
- Pregnancy or breastfeeding — no adequate human safety data exists for the injected or IV form.
- Cardiac history without prior discussion, or claustrophobia or limited time that makes a prolonged, symptomatic infusion risky.
- Anyone whose pharmacy cannot produce a current Certificate of Analysis for sterility and endotoxin testing on the specific lot being administered.
- Anyone being offered it specifically as an anti-aging or cognitive "cure," rather than as an unproven adjunct.
- Anyone expecting established dosing, frequency, or long-term safety data for repeated infusions — these have not been systematically studied.

Competitive athletes: NAD+ does not appear on the WADA Prohibited List and is not treated as a banned or monitored substance by USADA. The practical caveats for a competing athlete are not doping-status ones: contamination or mislabeling risk from an unregulated compounded product is real, and any multi-ingredient NAD+ IV formula should be checked against the current Prohibited List ingredient-by-ingredient, since NAD+'s own clean status doesn't extend to whatever else is in the bag.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller. That designation carries specific, enforceable obligations:

- A prescription in your name. Compounded from a prescription for one identified patient, after an evaluation.
- Licensed and inspected. A state licence, subject to inspection by the state board of pharmacy.
- Ingredients that can be traced. A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- Sterile preparation to a published standard. USP standards.

A paper trail. A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Not an approved medicine. FDA has twice declined to add it to a compounding bulks list on stability and toxicology grounds. Human outcome evidence for the injected form is limited. There is real science on NAD+ biology, far less on whether infusing it changes energy, cognition, ageing, or recovery — and a January 2026 warning letter documented three patients sickened by a contaminated infusion. Infusion speed matters: going too fast is the most common reason people feel unwell during a session, which is why infusions are given slowly.

Clinical detail

FDA has formally evaluated NAD+ for both the 503A and 503B compounding bulk-substance lists and declined both times, citing stability (pyrophosphate-linkage hydrolysis) and toxicology gaps; it currently sits in 503B "Category 1 — Under Evaluation." A January 2026 FDA warning letter to a compounding firm documented three patients with adverse reactions to an NAD+ infusion later found to contain bacterial endotoxin at 3,360 EU/mL, following the company's own Class I recall months earlier. FDA has separately warned the compounding industry generally that some compounders have used food-grade NAD+ — material intended for supplements, not sterile injection — to make IV products. Oral NAD+ precursors, such as nicotinamide riboside and nicotinamide mononucleotide, are regulated as dietary supplements under a different framework than the injected or IV form, which sits in the compounding-drug lane FDA has twice reviewed unfavorably.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment. NAD+ is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality. Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose. Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.

IMPORTANT SAFETY INFORMATION

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